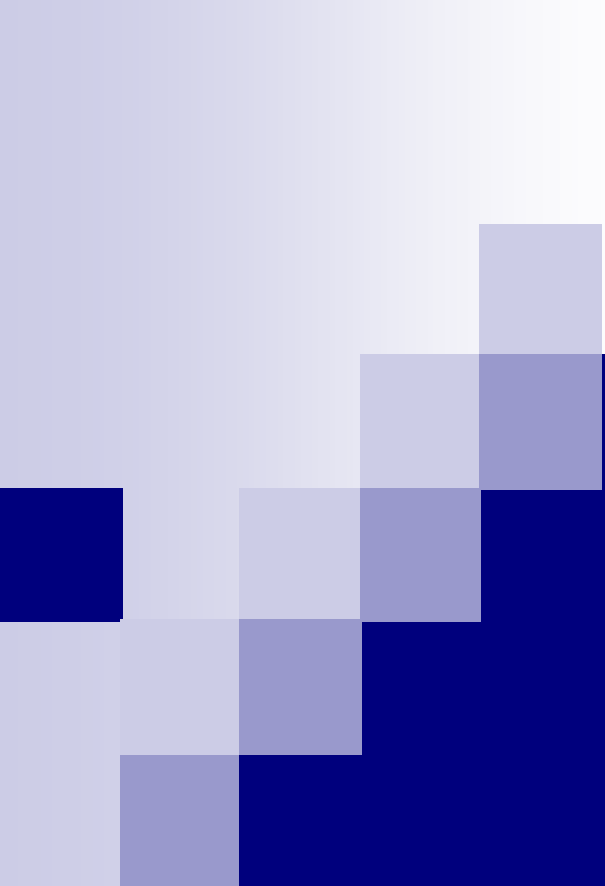




Software Defects - I



Lecture # 2

What is a Software Defect?

- A software defect is an error, flaw, mistake, failure, or fault in software that prevents it from behaving as intended (e.g., producing an incorrect or unexpected result)
- Software defects are also known as software errors or software bugs

Effects of Software Defects - 1

- Bugs can have a wide variety of effects, with varying levels of inconvenience to the user of the software. Some bugs have only a subtle effect on the program's functionality, and may thus lie undetected for a long time. More serious bugs may cause the software to crash or freeze leading to a denial of service

Effects of Software Defects - 2

- Others qualify as security bugs and might for example enable a malicious user to bypass access controls in order to obtain unauthorized privileges

Effects of Software Defects - 3

- The results of bugs may be extremely serious
- In 1996, the European Space Agency's US \$1 billion prototype **Arian 5** rocket was destroyed less than a minute after launch, due a bug in the on-board guidance computer program

Effects of Software Defects - 4

- In June 1994, a Royal Air Force **Chinook** crashed into the Mull of Kintyre, killing 29 people. An investigation uncovered sufficient evidence to convince that it may have been caused by a software bug in the aircraft's engine control computer

Effects of Software Defects - 5

- In 2002, a study commissioned by the US Department of Commerce' National Institute of Standards and Technology concluded that software bugs are so prevalent and detrimental that they cost the US economy an estimated US \$59 billion annually, or about 0.6 percent of the gross domestic product

Categories of Software Defects

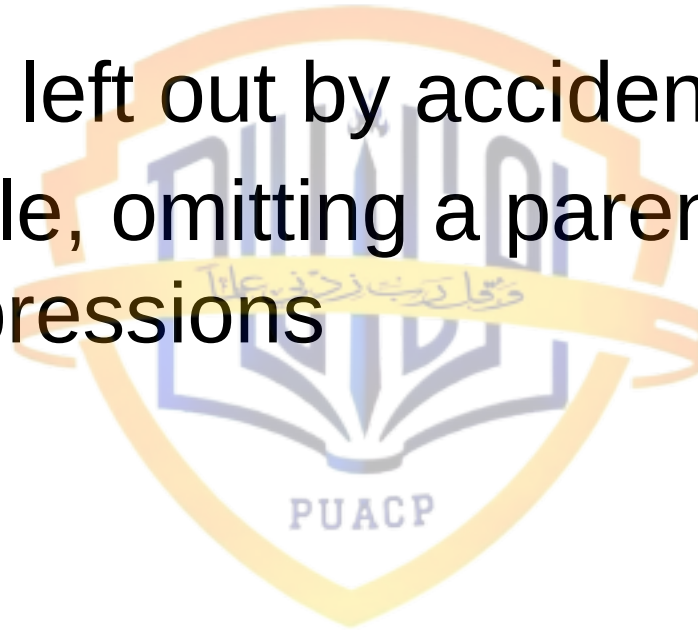
- Errors of commission
- Errors of omission
- Errors of clarity and ambiguity
- Errors of speed or capacity

Errors of Commission

- Something wrong is done
- A classic example at the code level would be going through a loop one time too many or branching on the wrong address

Errors of Omission

- Something left out by accident
- For example, omitting a parentheses in nested expressions



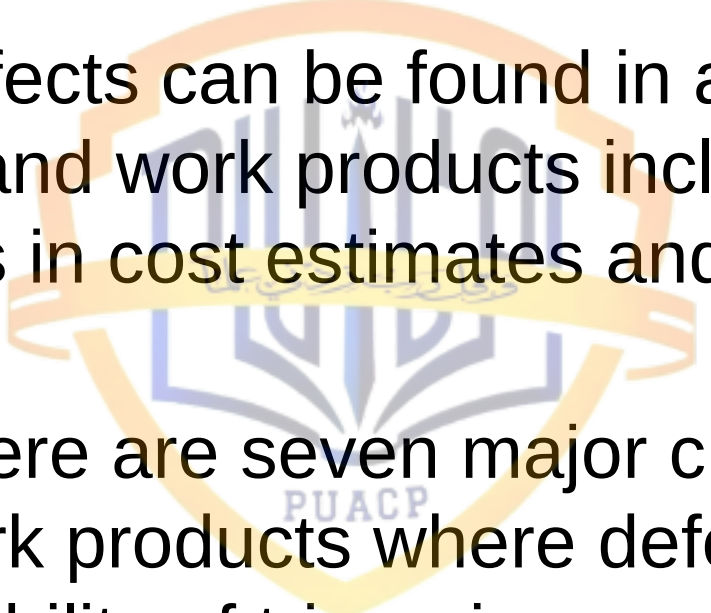
Errors of Clarity and Ambiguity

- Different interpretations of the same statement
- This kind of error is common with all natural language requirements and specification documents and user manuals, too.

Errors of Speed and Capacity

- Application works, but not fast enough

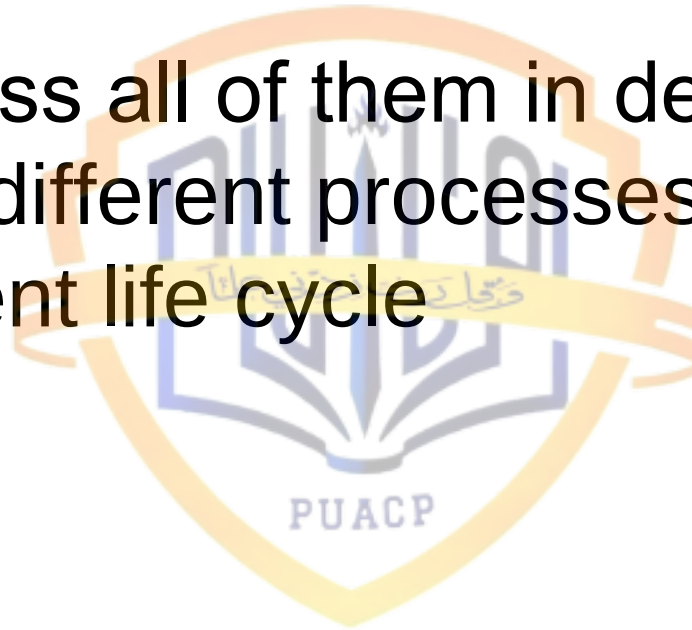


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- 
- Software defects can be found in any of the documents and work products including very serious ones in cost estimates and development plans
 - However, there are seven major classes of software work products where defects have a strong probability of triggering some kind of request for warranty repair if they reach the field

Software Defect Origins

- Errors in Requirements
- Errors in Design
- Errors in Source code
- Errors in User Documentation
- Errors due to “Bad fixes”
- Errors in Data and Tables
- Errors in Test Cases

- We'll discuss all of them in detail, when we talk about different processes of software development life cycle



Defect Discovery

- Defects are discovered by developers & testers (usually) before release
- Defects are discovered by customers and users (usually) after release
- Defects discovered after release can be embarrassing for the development team

Defect Discovery by Customers

- Rule 1: Defect discovery is directly related to the number of users
- Rule 2: Defect discovery is inversely related to the number of defects

Software Defect Elimination Strategies

- Effective defect prevention
- High levels of defect removal efficiency
- Accurate defect prediction before the project begins
- Accurate defect tracking during development
- Useful quality measurements
- Ensuring high levels of user-satisfaction

Defect Prevention and Removal

- Both defect prevention and removal techniques are used by the “best-in-the-class” companies
- Defect prevention is very difficult to understand, study, and quantify. We’ll talk about defect prevention in a later lecture
- Both non-test and testing defect removal techniques must be applied

Typical Defect Removal

- Inspections
 - Direct fault detection and removal
- Testing
 - Failure observation and fault removal



Inspections - 1

- Inspections are critical examinations of software artifacts by human inspectors aimed at discovering and fixing faults in the software systems

Inspections - 2

- Inspections are critical reading and analysis of software code or other software artifacts, such as designs, product specifications, test plans, etc
- Inspections are typically conducted by multiple human inspectors, through some coordination process. Multiple inspection phases or sessions may be used

Inspections - 3

- Faults are detected directly in inspection by human inspectors, either during their individual inspections or various types of group sessions
- Identified faults need to be removed as a result of the inspection process, and their removal also needs to be verified

Inspections - 4

- The inspection processes vary, but typically include some planning and follow-up activities in addition to the core inspection activity
- The formality and structure of inspections may vary, from very informal reviews and walkthroughs, to fairly formal variations of Fagan inspection, to correctness inspections approaching the rigor and formality of formal methods

Non-Test Defect Removal Methods

- Requirement inspections
- Design inspections
- Code inspections
- Test plan reviews
- Test-case inspections
- User documentation editing or reviews

Testing Defect Removal Methods

- Unit test by individual programmers
- New function testing
- Regression testing
- Performance testing
- Integration testing
- System testing
- Field test (external beta test)

Defect Removal

- Not all defects are equal when it comes to removal
- Requirements errors, design problems, and “bad fixes” are particularly difficult

Software Defect Origins & Defect Removal Effectiveness



The logo of the University of Peshawar Academic Center for Professional Studies (PUACP) is centered in the background. It features a shield with a book and a torch, with the acronym 'PUACP' at the bottom.

	Req. Defects	Design Defects	Code Defects	Document Defects	Perf. Defects
Reviews / Inspections	Fair	Excellent	Excellent	Good	Fair
Prototypes	Good	Fair	Fair	N/A	Good
Testing (all forms)	Poor	Poor	Good	Fair	Excellent
Correctness Proofs	Poor	Poor	Good	Fair	Poor

Defect Removal Efficiency

- Accumulation of defect statistics for errors found prior to delivery, and then for a predetermined period after deployment (usually one year)
- US averages: 85%
- Best projects in best US companies: 99%

Summary

- In today's lecture, we talked about software defects and where are they introduced in the software product
- We discussed the approaches to eliminating these defects

References

- Software Quality: Analysis and Guidelines for Success by Capers Jones
- Software Quality Engineering: Testing, Quality Assurance, and Quantifiable Improvement by Jeff Tian